HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed advanced environmental engineering scenarios for AI model training, focusing on contaminant transport and mass balance calculations within wastewater treatment processes.
- Authored rigorous, step-by-step technical solutions for complex environmental problems, including detailed chemical dosage calculations and hydraulic flow models, establishing gold-standard data for AI evaluation.
- Audited AI-generated environmental impact statements and remediation plans, meticulously evaluating technical accuracy, safety protocols, and compliance with EPA and ISO 14001 frameworks.
- Identified and provided structured feedback on logical errors in AI outputs, such as incorrect stoichiometry in biological processes or overlooked secondary environmental impacts, to improve model reasoning.
- Developed Python scripts to automate data ingestion and transformation for training datasets, processing approximately 5,000 records daily from diverse environmental monitoring sources.
- Implemented comprehensive data validation routines using SQL across PostgreSQL and Oracle databases, ensuring data integrity for over **100** distinct environmental datasets used in AI model development.
- Containerized data processing and AI model evaluation tools using Docker, facilitating consistent deployment and execution across development and staging environments on AWS EC2 instances.

SKILLS

Environmental Engineering – Contaminant Transport, Mass Balance, Wastewater Treatment, Aquatic Chemistry, Hydrology, Life Cycle Assessment (LCA), Hazardous Waste Remediation

AI & Machine Learning – AI Model Training, Data Annotation, Output Evaluation, Technical Review, Mathematical Reasoning, Regulatory Compliance, Stoichiometry

Programming & Scripting – Python, SQL

Databases – PostgreSQL, Oracle

Cloud & DevOps – AWS (EC2), Docker

Tools & Methodologies – EPA Tools, ISO 14001, Jenkins, Git, Jira, Confluence

PROJECTS

AI Training Data for Water Treatment Optimization

- Authored detailed ground-truth solutions for AI training, covering mass balance calculations and chemical dosage adjustments for simulated wastewater treatment plant scenarios.
- Developed a Python-based data generation script to create synthetic datasets simulating pollutant dispersion, ensuring adherence to EPA guidelines for AI model evaluation.
- Reviewed and annotated AI-generated reports on water quality impact assessments, identifying discrepancies in chemical equations and flow dynamics, and providing actionable feedback.

Environmental Compliance Scenario Authoring

- Designed complex environmental engineering scenarios focused on regulatory compliance, including hazardous waste remediation planning and documentation for ISO 14001 adherence.
- Wrote step-by-step technical explanations for AI reasoning, detailing environmental impact calculations and safety protocols for simulated remediation projects.
- Evaluated AI outputs for logical consistency in biological process stoichiometry and secondary environmental impact analysis, contributing to iterative model refinement.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)